BUSINESS SCHOOL

SKEMA

Derivatives and Fixed Income
PGE M1 Fall 2026

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Exam Topics

Forwards & Futures

Do you know / can you

- The difference between forwards & futures?
- How a consumer/producer would hedge using forwards?
- Compute the fair price of a forward?
  - How do storage costs / dividends & leasing affect the forward price?
- Construct an arbitrage strategy with forwards?
- Calculate the profit & loss for your forward position at the delivery date?
- How trading on margin & margin calls work, and the role of the clearinghouse?

Bonds

Do you know / can you

- What spot rates / zero rates are?
- Price a coupon-bond using spot rates or yield-to-maturity?
- When a bond trades at a premium, at par, at a discount?
- Know what a par yield is?
- Know the relation between spot rates, ytm and par yields depending on whether the yield curve is upward sloping, downward sloping, or flat, and how it relates to bonds trading at par vs at a discount?

Bonds



Do you know / can you

- The expected return of holding a bond from time 0 to time 1?
- What the forward rate is?
- How to interpret duration and convexity?
- How duration is broadly computed?
- How an interest rate swap works?
  - Who is hurt when the floating rate (SOFR) increases during the life of the swap?

Options

Do you know / can you

- The definition of each option
- The max profit & loss of each option, and the break-even price for positive profit?
- PCP? PCP with dividends? How to interpret individual option prices by re-arranging PCP?
- Construct arbitrage strategies?
- The lower arbitrage bound of a European call/put?
- When the value of a European call/put is equal to its lower bound?

Options

Do you know / can you

- The Black-Scholes formula?
- The assumptions underlying BS, and which ones are violated in real life?
- What implied volatility is?
- The differences between American and European options?
- The value of an American option relative to its European counterpart?
- The Greeks?
- How a straddle is constructed, and properties of the straddle?